

OTOC-NMR Experiment Error Mitigation, Modeling & Results

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Paper: arXiv:2510.19550 (2025)

See also OTOC2: Nature v. 646, p. 825 (2025)
Google

The OTOC-NMR team

QC Experiment



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N. Noll



D. Kafri

Numerical modeling

Error mitigation



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M. Claassen



T. Schuster

OTOC theory



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Hardware and Pathfinding support



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J. Gross



A. Grajales



C. Quintana



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A. Ajoy



C. Bengs



C. Ramanathan

Chemistry modeling and algorithms



A. White



T. O'Brien



N. Rubin



R. Babbush

Google Deepmind



J. Bausch



S. Shirobokov



A. Senior



N. Tomasev

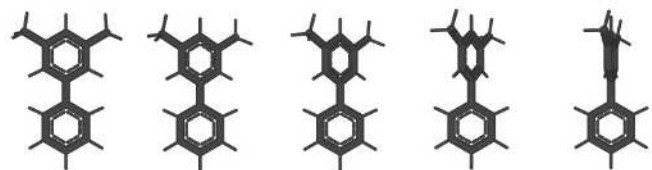


F. Heras

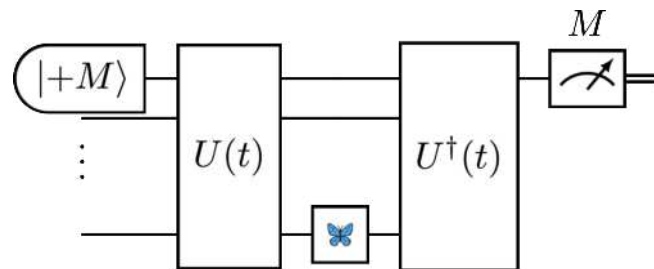


B. Kovlovskii

Noiseless Benchmarks and Experimental Targets

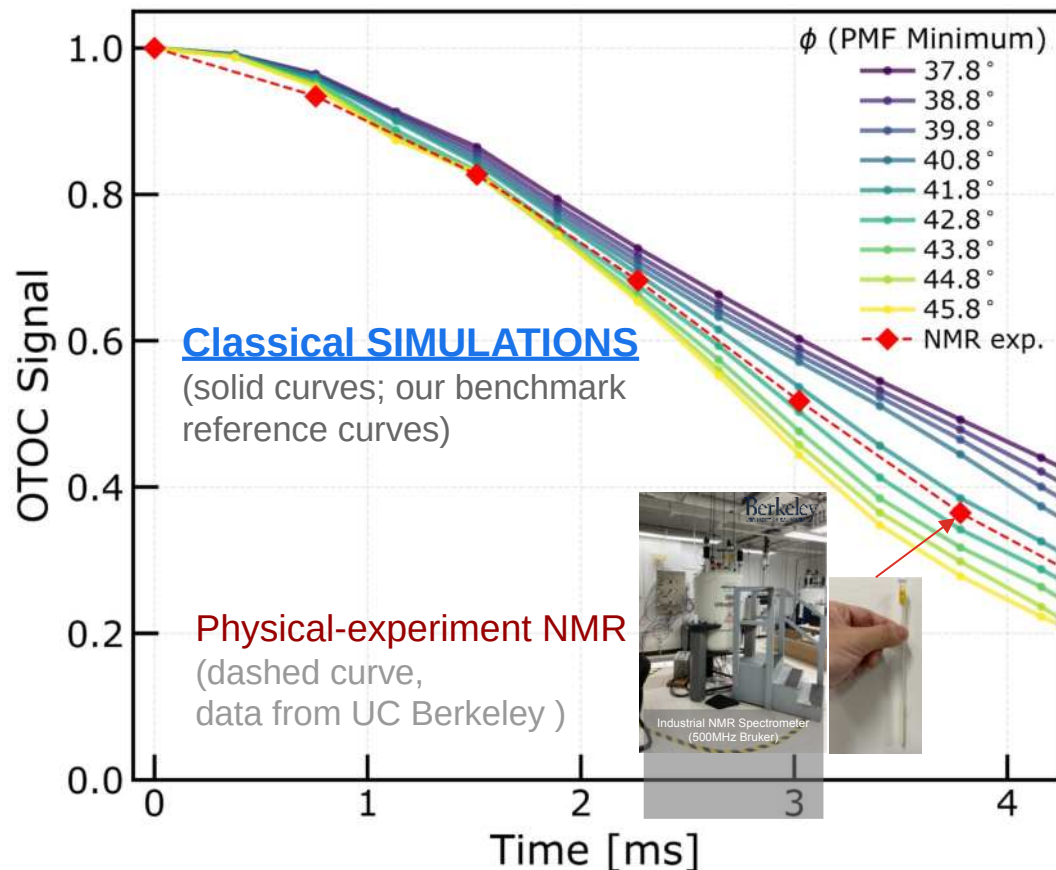


$$\{\hat{H}_{\text{trial},i}\}$$



Most of the signal is at $t > 3$ ms

Target $\langle \text{OTOC} \rangle$ with avg. $\varepsilon \approx 0.04$



Willow 105 Experiment: Partial Mitigation

Optimized cals

DD

Light-cone filter
(double-sided)

Readout mitigation
(real time)

Twirling 2Q gates
(Clifford SQ Sub)

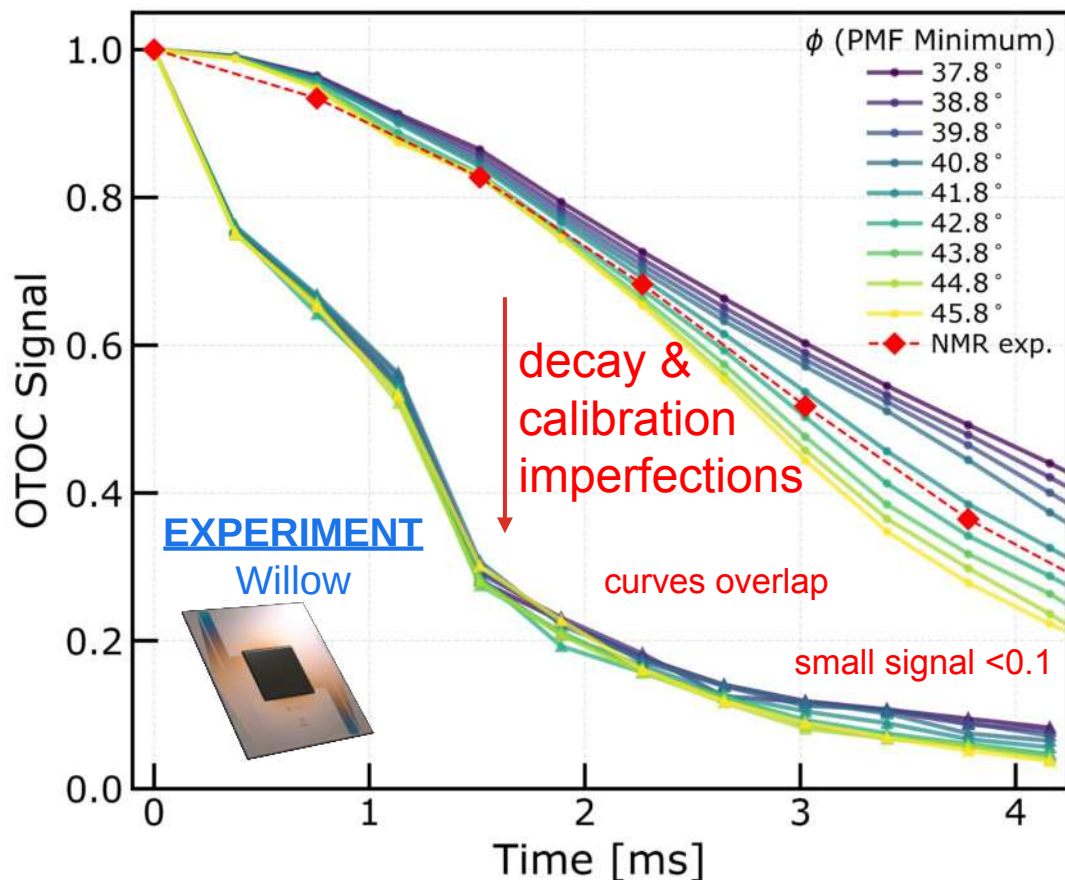
Twirling idle
(Clifford SQ)

We need to distinguish from these curves the correct one and find noise-free answers for $\langle \text{OTOC} \rangle$ to $\varepsilon \approx 0.04$

Data from
Willow 105, Canyon, Rodrigo Cor.

Cost per point: 300 circuits $\times$ 10K shots

Cost per curve: 3×10^6 shots

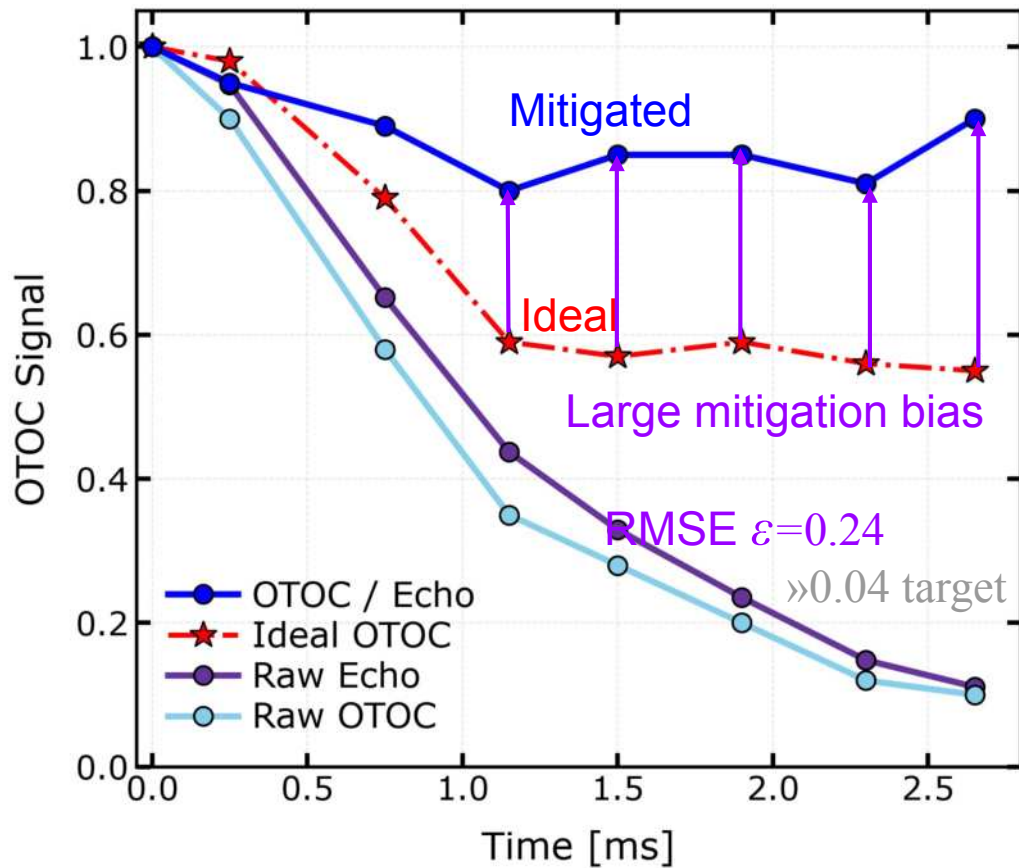


Initial Mitigation 1: Echo Rescaling

Use LS echo (no butterfly) signal size to rescale OTOC, employed previously*

$$C_{\text{Mitigated}}(t) = \frac{C_{\text{OTOC}}(t)}{C_{\text{Echo}}(t)}$$

Toluene molecule experiment (fSim)
Willow 105, Redwood, Amir K.



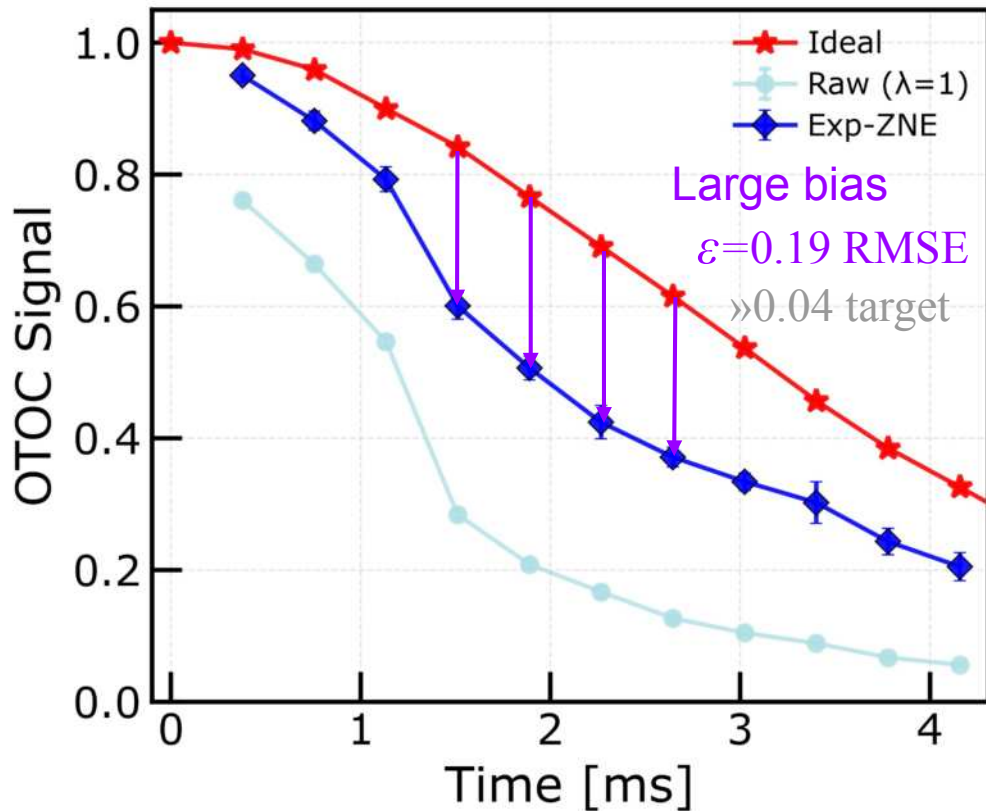
Initial Mitigation 2: Standard Exponential ZNE

Alternative Methods Evaluated

PEC†	Sampling cost prohibitive
PM-PEC	Too much bias
ZNE*	Standard ZNE shown here

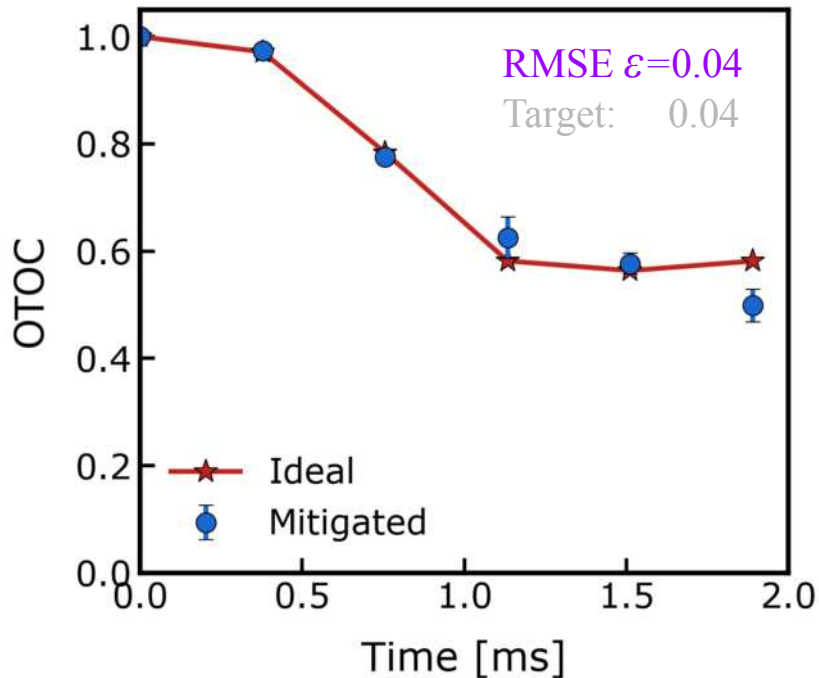
We tested exponential (exp), poly, and linear models. Exp-ZNE performed better than others.

DBMP molecule experiment (CZ)
Willow 105, Canyon, Rodrigo C.

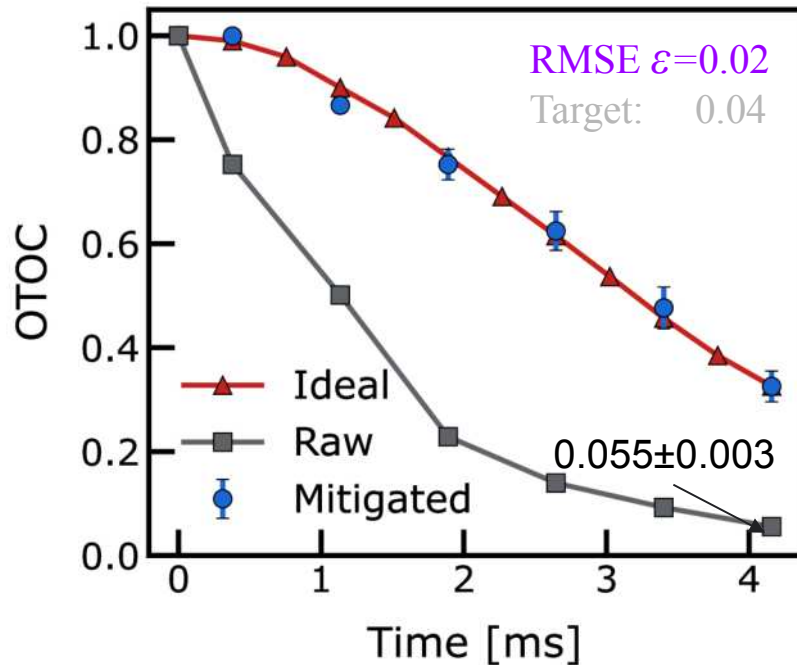


Mitigation using Pauli Path ZNE (to be introduced in rest of talk)

Toluene: fSim, Trotter



DMBP: CZ, AlphaEvolve



We observed overall comparable performance over different experimental runs on different days.

Optimized
cals

DD

Light-cone filter
(double-sided)

Readout mitigation
(real time, T-REX)

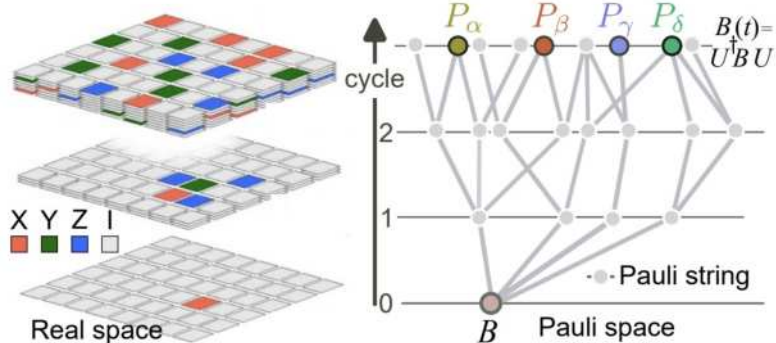
Twirling 2Q gates
(Clifford SQ Sub)

Twirling idle
(Clifford SQ)

Can a path-based unraveling of noisy evolution expose structural properties to enable more accurate error mitigation?

Pauli paths

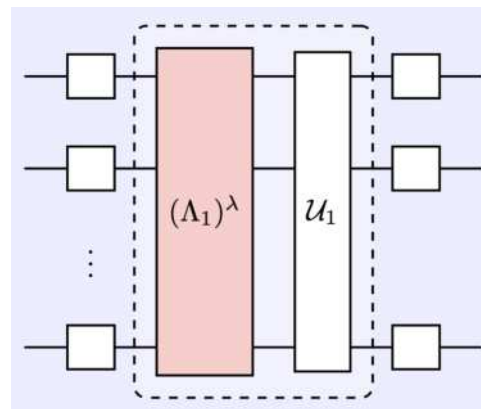
(we will generalize Pauli paths)



<https://arxiv.org/pdf/2506.10191>

+

Noise Response & Error Mitigation



T134-v2 fSim and Canonical Gates - Twirling and Folding (Zlatko).pdf

See Aharonov *et al.* (2023) for foundations;
Begušić *et al.* (2024) and $\dagger$ Rudolph *et al.* (2025) for frameworks;
Angrisani *et al.* (2025) and Schuster *et al.* (2025) for general noise.

The Laplace Structure of Circuit Response in NMR OTOC

$$C(\lambda) \approx \int_0^\infty \underbrace{e^{-\lambda\gamma w}}_{\text{Noise}} \underbrace{R(w)}_{\text{Reactivity}} dw$$

An effective **2 parameter model**:

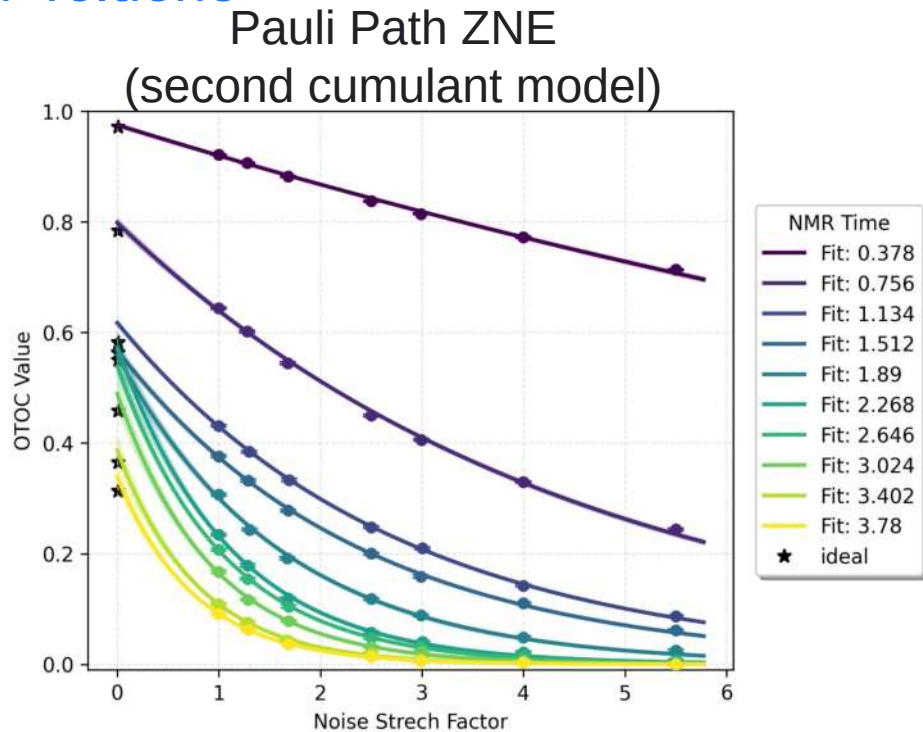
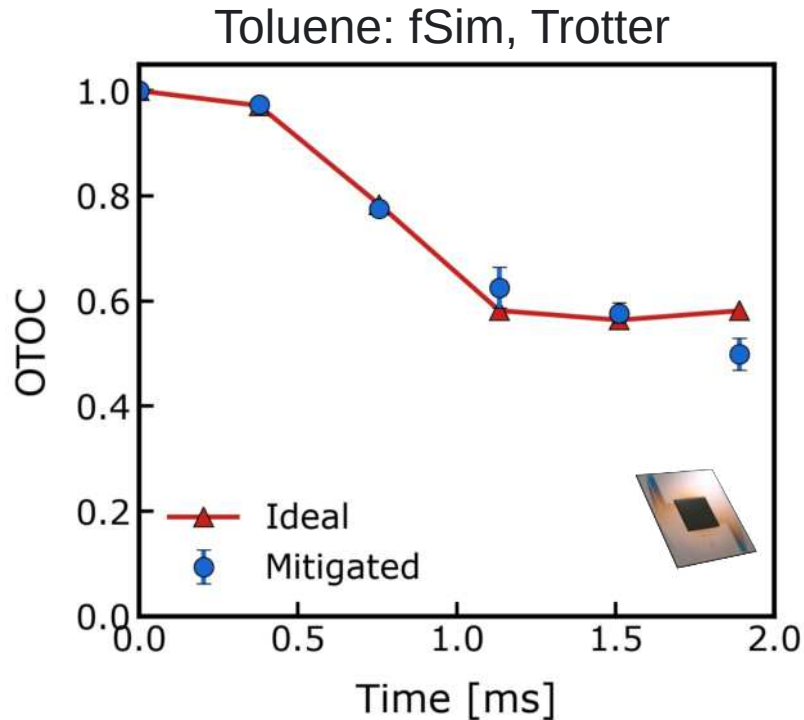
$$C(\lambda; \bar{H}, \beta) = C_{\text{noise-free}} e^{-\bar{H}\lambda} e^{\frac{1}{2}\bar{H}^2\beta^2\lambda^2} \frac{1 + \operatorname{erf}\left(\frac{1-\beta^2\bar{H}\lambda}{\beta\sqrt{2}}\right)}{1 + \operatorname{erf}\left(\frac{1}{\beta\sqrt{2}}\right)}$$

Relative width of the Pauli path Hamming distribution (sigma / Hbar)

Average Pauli Path Hamming weight x processor damping

Warning: Fitting this is sensitive!
We use Bayesian time-correlated model.

Mitigation using Pauli Path ZNE for Toluene



We observed overall comparable performance over different experimental runs on different days.

Optimized
cals

DD

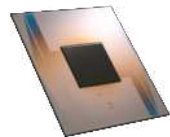
Light-cone filter
(double-sided)

Readout mitigation
(real time)

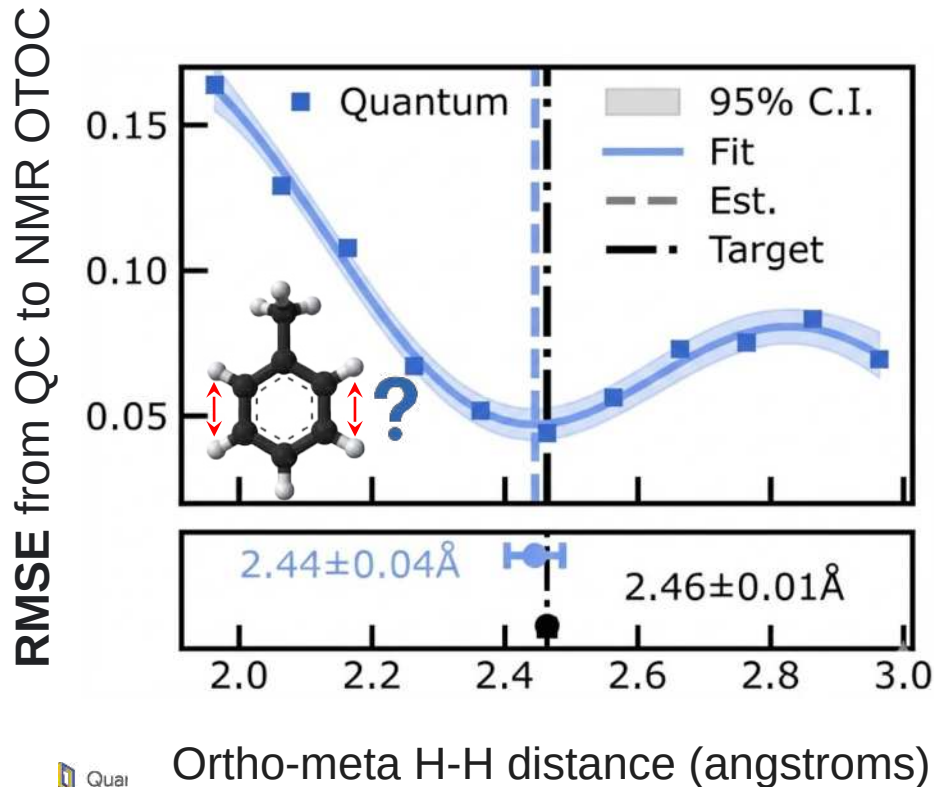
Twirling 2Q gates
(Clifford SQ Sub)

Twirling idle
(Clifford SQ)

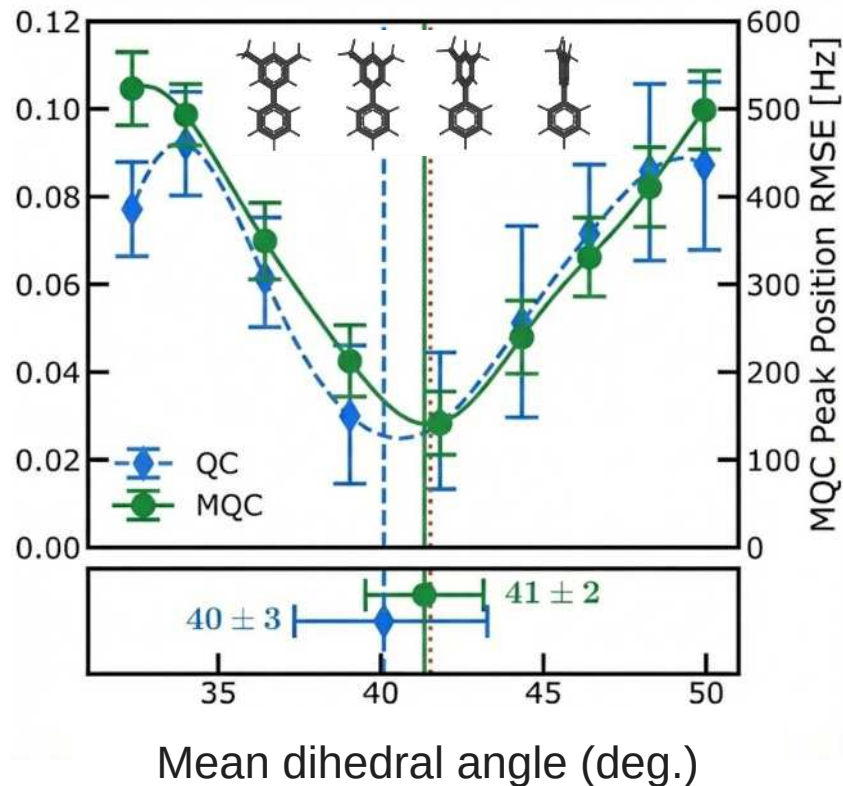
Pauli Path ZNE recovers physical NMR benchmarks



Toluene (fSim, Trotter)



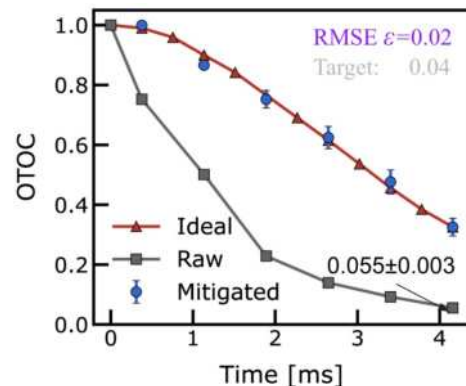
DMBP (CZ, AlphaEvolve)



Conclusions

I. Performance on Hardware for Pauli Path ZNE

- **Pauli Path ZNE** exceeds performance of other mitigation strategies in NMR OTOC, achieving consistently low RMSE < 0.04 .
- **Broadly useful:** Across fSim, CZ, cPHASE gates; Preliminaries suggest direct reuse across other application regimes.
- **Agrees with gold standard:** Mitigation helps us achieve accuracy and (Angstroms and ~ 0.05 angular rads) that is similar to independent NMR spectroscopic measurements of molecular structure.



II. Pauli Path ZNE (Manuscript Soon)

- **See also ML for Error Mitigation**
[See my A07 invited talk on Nature Machine Intelligence v6, 1478 (2024)]



6 mo. Internship? FTE Roles?

Google Quantum AI

Contact me

For more about our
presence at APS



Google Quantum AI

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